



TRAINING PROGRAMME

SOC Analyst (CCNA Cybersecurity)

Monitor, detect, investigate and respond in a security operations centre

57 HOURS

FOR IT and network staff moving into security operations

PREPARES FOR Cisco CCNA Cybersecurity (200-201 CBROPS v1.2)



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Overview

This programme prepares participants for the work of a security operations centre analyst and for the Cisco CCNA Cybersecurity exam (200-201 CBROPS). They learn how a SOC works, analyse Windows and Linux systems, review network protocols and security infrastructure from the analyst's side, monitor traffic, apply cryptography, protect endpoints and analyse malware. They then work with security data in a SIEM, detect attacks with MITRE ATT&CK and Sigma rules, use threat intelligence and AI-assisted triage, investigate incidents following NIST SP 800-61 Rev. 3 and apply security procedures. Every chapter ends with a lab, and the programme closes with a final project: a SOC simulation on a captured attack.

At a glance

#	PHASE	FOCUS	HOURS
01	SOC Foundations & Operating Systems	The SOC, security concepts, Windows and Linux for analysts	9 h
02	Network Security & Monitoring	Protocols, security infrastructure, attacks, tcpdump, Wireshark, NetFlow	9 h
03	Cryptography, Endpoints & Malware	Cryptography and PKI, endpoint protection, vulnerabilities, malware analysis	9 h
04	Detection & Threat Intelligence	SIEM and SOAR, Security Onion, MITRE ATT&CK, Sigma, threat intelligence, AI	9 h
05	Incident Response & Policies	NIST SP 800-61 Rev. 3, evidence, kill chain, diamond model, playbooks	6 h
06	Final Project	A SOC simulation on a captured attack	6 h
	Buffer	Catch-up, revision, extra lab time	9 h
	Total		57 h

Key facts

AUDIENCE	IT and network staff moving into security operations
PREREQUISITES	Networking basics (course: Introduction to Networks (CCNA 1), or Networking for Cloud & DevOps); Linux command line helps. Networking is recapped, not taught
DURATION	57 h: 48 h of chapters + 9 h buffer
FORMAT	Short lectures followed by guided labs on captured traffic and logs; one attack scenario followed through the course
LAB ENVIRONMENT	One laptop per participant (16 GB RAM) with an analyst workstation and Kali Linux VMs prepared by the trainer; one Security Onion server per team (4 cores, 24 GB RAM, 200 GB disk: the Security Onion standalone minimum), provided by the organiser; isolated lab network; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final SOC simulation with an incident report; certificate of completion for participants who complete both

OUTCOME

Participants can triage alerts, investigate network and host evidence, map an attack to MITRE ATT&CK, write an incident report and sit the 200-201 CBROPS exam

PHASE 01

9 H

SOC Foundations & Operating Systems

Participants learn what a SOC does, how its tiers work and the security concepts analysts use every day, then examine Windows and Linux from the analyst's side: processes, registry, services, permissions and the logs that record what happened. Validated lab: a suspicious process and the user who started it traced through Windows event logs and Linux logs.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	The SOC & Threat Landscape	Threats, cybercrime, SOC roles and tiers	Lecture + Workshop	1.5 h
1.2	Security Concepts	CIA triad, defence in depth, access control models, CVSS, risk, attack surface, deterministic vs probabilistic analysis	Lecture + Workshop	1.5 h
1.3	Windows for Analysts	Architecture, processes, registry, event logs, security tools	Lecture + Lab	3 h
1.4	Linux for Analysts	Logs, permissions, services, hardening	Lecture + Lab	3 h

PHASE 02

9 H

Network Security & Monitoring

Participants review the protocols an analyst reads every day, see how firewalls, IPS and proxies protect the network, study the common attacks on networks and applications, then capture and analyse traffic with tcpdump, Wireshark and NetFlow. Validated lab: a packet capture in which each participant identifies a scan, a brute-force attempt and data exfiltration.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Network Protocols Review	Protocols seen from the analyst's side (recap of Introduction to Networks or Networking for Cloud & DevOps)	Lecture + Lab	1.5 h
2.2	Network Security Infrastructure	Firewalls, IPS, proxies, segmentation	Lecture + Workshop	1.5 h
2.3	Attacks on Networks & Applications	Tools, common threats and attacks	Lecture + Lab	3 h
2.4	Network Monitoring	tcpdump, Wireshark, NetFlow, attacks on protocols and services, data types (full packet capture, session, transaction, statistical, metadata, alert), 5-tuple, regular expressions	Lecture + Lab	3 h

PHASE 03

9 H

Cryptography, Endpoints & Malware

Participants see how cryptography and certificates protect data and how attackers hide in encrypted traffic, analyse endpoints and assess vulnerabilities, then examine malware safely with hashes, static analysis and a sandbox to extract indicators of compromise. Validated lab: a malware sample analysed in a sandbox, its indicators of compromise extracted and searched for in the endpoint logs.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Cryptography & PKI	Integrity, confidentiality, certificates	Lecture + Lab	3 h
3.2	Endpoint Protection & Vulnerabilities	EDR, host-based analysis, vulnerability assessment	Lecture + Lab	3 h
3.3	Malware Analysis Basics	Hashes, static and sandbox analysis, indicators of compromise	Lecture + Lab	3 h

PHASE 04

9 H

Detection & Threat Intelligence

Participants bring log sources into a SIEM, work in Security Onion, triage alerts as true or false positives, map attacks to MITRE ATT&CK and write Sigma detection rules, enrich alerts with threat intelligence, and see where AI-assisted triage helps and where it adds risk. Validated lab: a Sigma rule written for an attack technique, converted for the SIEM and tested against the course attack.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Security Data & SIEM	Log sources, SIEM and SOAR, Security Onion (Elastic, Zeek, Suricata), other SIEMs such as Wazuh	Lecture + Lab	3 h
4.2	Alert Analysis & Detection	MITRE ATT&CK, Sigma rules, true and false positives	Lecture + Lab	3 h
4.3	Threat Intelligence	Sources, sharing, frameworks	Lecture + Lab	1.5 h
4.4	AI in Security Operations	AI-assisted triage and detection, risks	Lecture + Workshop	1.5 h

PHASE 05

6 H

Incident Response & Policies

Participants investigate incidents following NIST SP 800-61 Rev. 3, handle evidence so it stays usable, reconstruct attacks with the kill chain and the diamond model, then work with the playbooks, asset management and compliance rules that frame SOC work. Validated lab: an incident timeline and a playbook step completed for the course attack.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Forensics & Incident Response	NIST SP 800-61 Rev. 3, evidence handling, kill chain, diamond model, evidence types and chain of custody	Lecture + Lab	3 h
5.2	Security Policies & Procedures	Incident handling playbooks, asset management, compliance (PCI DSS, HIPAA, SOX), CSIRT/PSIRT/CERT, SOC metrics (time to detect, contain, respond)	Lecture + Workshop	3 h

PHASE 06

6 H

Final Project

Teams of two or three work a shift in a simulated SOC: they investigate the alerts of a captured attack, reusing the phase labs, then brief the group as they would a SOC manager.

#	CHAPTER	CONTENT	FORMAT	HOURS
6.1	Triage & Investigation	Triage the alerts in Security Onion, investigate network and host evidence	Project	3 h
6.2	Timeline & Report	Attack timeline mapped to MITRE ATT&CK, indicators of compromise, incident report	Project	1.5 h
6.3	Debrief & Review	Present the findings and recommendations; questions and feedback	Project	1.5 h

FINAL PROJECT

SOC Simulation

The alerts of a captured attack investigated end to end: triage of true and false positives, network and host evidence, an attack timeline mapped to MITRE ATT&CK, indicators of compromise and an incident report with recommendations.

STACK Security Onion (Elastic, Zeek, Suricata), Wireshark, Sigma, MITRE ATT&CK Navigator

DELIVERABLE Incident report with timeline and indicators, debrief presentation